

Soliton solution for nonlinear partial differential equations by cosine-function method

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Abstract

In this Letter, we established a traveling wave solution by using Cosine-function algorithm for nonlinear partial differential equations. The method is used to obtain the exact solutions for five different types of nonlinear partial differential equations such as, general equal width wave equation (GEWE), general regularized long wave equation (GRLW), general Korteweg–de Vries equation (GKdV), general improved Korteweg–de Vries equation (GIKdV), and Coupled equal width wave equations (CEWE), which are the important soliton equations.

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1. Introduction

The study of numerical methods for the solution of nonlinear partial differential equations has enjoyed an intense period of activity over the last 40 years from both theoretical and practical points of view. Improvements in numerical techniques, together with the rapid advances in computer technology, have meant that many of the partial differential equations arising from engineering and scientific applications, which were previously intractable, can now, be routinely solved [1]. Finite difference methods approximate the differential operators and hence, solve the difference equations. In finite element method the continuous domain is represented as a collection of a finite number N of subdomains known as elements. The collection of elements is called the finite element mesh. For time dependent problems, the differential equations are approximated by the finite element method to obtain a set of ordinary differential equations in time. These differential equations are solved approximately by finite difference methods or other methods. In all finite difference and finite elements it is necessary to have a boundary and initial conditions. But the Adomian decomposition method, which has been developed by George Adomian (Solving Frontier Problem of Physics: The Decomposition Method, Kluwer Academic Publishers, Boston, MA, 1994) depends only on the initial conditions to obtain solution in series form which almost converges to the exact solutions of the problem. In recently years, some other ansatz methods have been developed, such as, the tanh method [2,3], the extended tanh-function method [4,5], the modified extended tanh-function method [6–11], variational iteration method [12–17], the generalized hyperbolic-function [18,19], the separation of variables method [20,21] and the sine–cosine method [22,23].

The aim of this Letter is extended the cosine function-method to find the exact solutions of some important nonlinear partial differential equations such as the general equal width wave (GEW) equation, the general Regularized long wave (GRLW) equation,

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the general Korteweg–de Vries (GKdV) equation, the general improved Korteweg–de Vries (GIKdV) equation and the coupled equal width wave equations, which are the five of important soliton equations.

2. The cosine function method

Consider the nonlinear partial differential equation in the form

$$F(u, u_t, u_x, u_{xx}, u_{xxt}, \dots) = 0, \quad (1)$$

where $u(x, t)$ is the solution of nonlinear partial differential equation Eq. (1). We use the transformations,

$$u(x, t) = f(\xi), \quad (2)$$

where $\xi = x - ct$. This enables us to use the following changes:

$$\frac{\partial}{\partial t}(\cdot) = -c \frac{d}{d\xi}(\cdot), \quad \frac{\partial}{\partial x}(\cdot) = \frac{d}{d\xi}(\cdot), \quad \frac{\partial^2}{\partial x^2}(\cdot) = \frac{d^2}{d\xi^2}(\cdot), \quad \dots \quad (3)$$

Using Eq. (3) to transfer the nonlinear partial differential equation Eq. (1) to nonlinear ordinary differential equation

$$G(f, f', f'', f''', \dots) = 0. \quad (4)$$

The solution of Eq. (4) can be expressed in the form:

$$f(\xi) = \lambda \cos^\beta(\mu\xi), \quad |\xi| \leq \frac{\pi}{2\mu}, \quad (5)$$

where λ , β and μ are unknown parameters which will be determined. Then we have:

$$\begin{aligned} f' &= \frac{df(\xi)}{d\xi} = -\lambda\beta\mu \cos^{\beta-1}(\mu\xi) \sin(\mu\xi), \\ f'' &= \frac{d^2f(\xi)}{d\xi^2} = -\lambda\beta\mu^2 \cos^\beta(\mu\xi) + \lambda\mu^2\beta(\beta-1) \cos^{\beta-2}(\mu\xi) - \lambda\mu^2\beta(\beta-1) \cos^\beta(\mu\xi). \end{aligned} \quad (6)$$

Substituting Eq. (6) into the nonlinear ordinary differential equation Eq. (4) gives a trigonometric equation of $\cos^\alpha(\mu\xi)$ terms. To determine the parameters first balancing the exponents of each pair of cosine to determine α . Then we collect all terms with the same power in $\cos^\beta(\mu\xi)$ and put to zero their coefficients to get a system of algebraic equations among the unknowns β , λ and μ . Now, the problem is reduced to a system of algebraic equations that can be solved to obtain the unknown parameters β , λ and μ . Hence, the solution considered in Eq. (5) is obtained. The above analysis yields the following theorem:

Theorem. *The exact analytical solution of the nonlinear partial differential equations Eq. (1) can be determined in the form Eq. (5) where all constants found from the algebraic equations after its solutions.*

3. Applications

In order to illustrate the effectiveness of the proposed method five examples in mathematical are chosen as follows

3.1. Test problem

3.1.1. The general equal width wave equation (GEWE)

Consider the following problem: Find functions $u(x, t)$ satisfying the general equal width wave equation (GEWE) [24–27] in the form,

$$u_t + \varepsilon u^p u_x - \nu u_{xxt} = 0. \quad (7)$$

By using the wave variable $\xi = x - ct$ and $u(x, t) = f(\xi)$ then GEWE equation becomes

$$-c \frac{df(\xi)}{d\xi} + \varepsilon f^p(\xi) \frac{df(\xi)}{d\xi} + \nu \frac{d^3f(\xi)}{d\xi^3} = 0. \quad (8)$$

Integrating Eq. (8) gives

$$-cf(\xi) + \frac{\varepsilon}{p+1} (f(\xi))^{p+1} + \nu \frac{d^2f(\xi)}{d\xi^2} = 0. \quad (9)$$

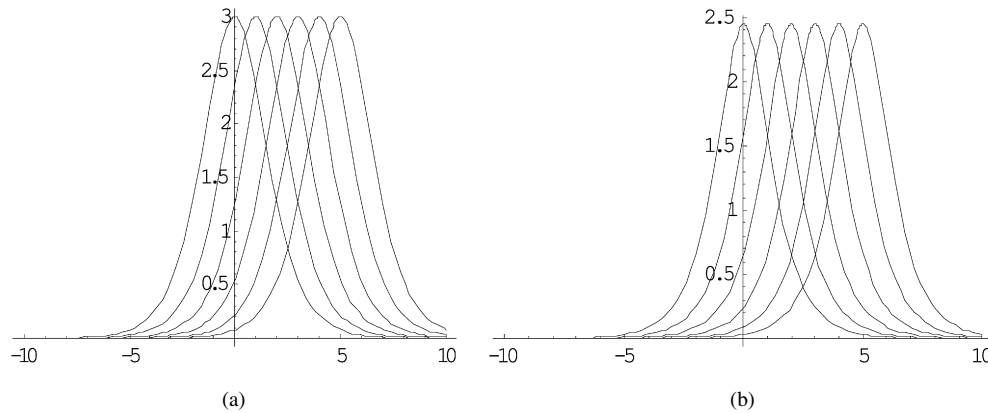


Fig. 1. (a) Soliton solution for EWE; (b) soliton solution for MEWE.

Table 1

The conservative quantities the EWE, MEWE and GEWE equations with increase time

T	p	I_1			I_2		
		1	2	3	1	2	3
0		12	7.6953	6.04182	28.8	16	11.4365
1		12	7.6953	6.04182	28.8	16	11.4365

Substituting Eq. (6) into (9) gives:

$$-c\lambda \cos^\beta(\mu\xi) + \frac{\varepsilon\lambda^{p+1}}{p+1} \cos^{(p+1)\beta}(\mu\xi) - c\nu\lambda\beta\mu^2 \cos^\beta(\mu\xi) + \nu c\lambda\beta(\beta-1)\mu^2 \cos^{\beta-2}(\mu\xi) - c\nu\lambda\beta(\beta-1)\mu^2 \cos^\beta(\mu\xi) = 0. \quad (10)$$

The last equation is satisfied only if the following system of algebraic equations holds:

$$\begin{aligned} -c\lambda - c\lambda\nu\beta\mu^2 - c\lambda\mu^2\nu\beta(\beta-1) &= 0, & (p+1)\beta &= \beta-2, \\ \frac{\varepsilon\lambda^{p+1}}{p+1} + c\lambda\mu^2\nu\beta(\beta-1) &= 0, \end{aligned} \quad (11)$$

using MATHEMATICA package software for solving the system Eq. (11) we obtain:

$$\beta = \frac{-2}{p}, \quad \mu = \pm \frac{ip}{2\sqrt{\nu}} \quad \text{and} \quad \lambda = 2^{-1/p} \left(\frac{c(2+3p+p^2)}{\varepsilon} \right)^{1/p}, \quad (12)$$

substituting Eq. (12) into Eq. (5) we obtain the exact soliton solution of the GEWE equation in the form

$$u(x, t) = 2^{-1/p} \left(\frac{c(2+3p+p^2)}{\varepsilon} \right)^{1/p} \cos^{-2/p} \left(\frac{\pm ip}{2\sqrt{\nu}} (x - ct) \right). \quad (13)$$

As a special case when $p = 1$ gives the soliton solution of the EWE equation as in [24], when $p = 2$ gives the soliton solution of the MEWE equation as in [29] and also we can get the soliton solution of the GEWE equation as in [30]. Fig. 1 shows the soliton solution for the EWE, MEWE and GEWE equations with increase time. Also, we can calculate the conservative quantities when $p = 1$ and 2 as given in Table 1. It is to be noted that the obtained results are quite good with that obtained [28–30].

3.1.2. The general regularized long wave (GRLW) equation

We consider the general regularized long wave equation for long waves in shallow water written as [31–35]:

$$u_t + u_x + \varepsilon u^p u_x - \nu u_{xxt} = 0. \quad (14)$$

Using the transformation $u(x, t) = f(\xi)$ where $\xi = x - ct$, then the GRLW equation becomes

$$-c \frac{df(\xi)}{d\xi} + \frac{df(\xi)}{d\xi} + \varepsilon f^p(\xi) \frac{df(\xi)}{d\xi} + c\nu \frac{d^3 f(\xi)}{d\xi^3} = 0. \quad (15)$$

Integrating Eq. (15) gives

$$(1-c)f(\xi) + \frac{\varepsilon}{p+1} (f(\xi))^{p+1} + c\nu \frac{d^2 f(\xi)}{d\xi^2} = 0. \quad (16)$$

Substituting Eq. (6) into Eq. (16) becomes

$$(1-c)\lambda \cos^\beta(\mu\xi) + \frac{\varepsilon\lambda^{p+1}}{p+1} \cos^{(p+1)\beta}(\mu\xi) - cv\lambda\beta\mu^2 \cos^\beta(\mu\xi) + cv\lambda\beta(\beta-1)\mu^2 \cos^{\beta-2}(\mu\xi) - cv\lambda\beta(\beta-1)\mu^2 \cos^\beta(\mu\xi) = 0. \quad (17)$$

By equating the exponents and the coefficients of each pair of the cosine function we obtain the following system of algebraic equations:

$$(1-c)\lambda - cv\lambda\beta\mu^2 - cv\lambda\mu^2\beta(\beta-1) = 0, \quad (p+1)\beta = \beta-2, \\ \frac{\varepsilon\lambda^{p+1}}{p+1} + cv\lambda\mu^2\beta(\beta-1) = 0. \quad (18)$$

By using MATHEMATICA package software for solving the system Eq. (18) we get:

$$\beta = \frac{-2}{p}, \quad \mu = \pm \frac{\sqrt{1-c}p}{2\sqrt{vc}} \quad \text{and} \quad \lambda = 2^{-1/p} \left(\frac{-2+2c-3p+3cp-p^2+cp^2}{\varepsilon} \right)^{1/p}. \quad (19)$$

Then by substituting Eq. (19) into Eq. (5) then, the exact soliton solution of the GRLW equation can be written in the form:

$$u(x, t) = 2^{-1/p} \left(\frac{-2+2c-3p+3cp-p^2+cp^2}{\varepsilon} \right)^{1/p} \cos^{-2/p} \left(\pm \frac{\sqrt{1-c}p}{2\sqrt{vc}}(x-ct) \right). \quad (20)$$

3.1.3. The GKdV and GIKdV equations

The general Korteweg–de Vries (GKdV) equation [36,37], for long waves in shallow water has the form:

$$u_t + \varepsilon u^p u_x + \gamma u_{xxx} = 0. \quad (21)$$

In the same manner to solve Eq. (21) by cosine-function method, we obtain the following system of equations as follows:

$$-c\lambda - \gamma\lambda\beta\mu^2 - \gamma\lambda\mu^2\beta(\beta-1) = 0, \quad (p+1)\beta = \beta-2, \\ \frac{\varepsilon\lambda^{p+1}}{p+1} + \gamma\mu^2\beta(\beta-1) = 0. \quad (22)$$

This can be solved using MATHEMATICA package software to get

$$\beta = \frac{-2}{p}, \quad \mu = \pm \frac{i\sqrt{c}p}{2\sqrt{\gamma}}, \quad \lambda = 2^{-1/p} \left(\frac{c(2+3p+p^2)}{\varepsilon} \right)^{1/p}. \quad (23)$$

So the exact soliton solution of GKdV equation can be written in the form

$$u(x, t) = 2^{-1/p} \left(\frac{c(2+3p+p^2)}{\varepsilon} \right)^{1/p} \cos^{-2/p} \left(\pm \frac{i\sqrt{c}p}{2\sqrt{\gamma}}(x-ct) \right). \quad (24)$$

In a similar way, when we consider the general improved Korteweg–de Vries (GIKdV) equation in the form [27,37]

$$u_t + \varepsilon u^p u_x + \gamma u_{xxx} - \nu u_{xxt} = 0. \quad (25)$$

The exact soliton solution of this soliton equation can be deduced to the form:

$$u(x, t) = 2^{-1/p} \left(\frac{c(2+3p+p^2)}{\varepsilon} \right)^{1/p} \cos^{-2/p} \left(\pm \frac{i\sqrt{c}p}{2\sqrt{\gamma+cv}}(x-ct) \right). \quad (26)$$

3.1.4. Coupled equal width wave equation (CEWE)

In this case we consider the coupled equal width wave equations, in the normalized form

$$u_t + uu_x - u_{xxt} + vv_x = 0, \quad v_t + vv_x - v_{xxt} = 0, \quad (27)$$

where the subscripts t and x denoting to the differentiation with respect to time and space respectively, the boundary conditions $u, u', u'' \rightarrow c_1$ and $v, v', v'' \rightarrow c_2$ as $x \rightarrow \pm\infty$ such that c_1, c_2 are arbitrary constants. Suppose $u(x, t) = f(\xi)$ and $v(x, t) = g(\xi)$ then the coupled EWE becomes

$$-c \frac{df(\xi)}{d\xi} + f(\xi) \frac{df(\xi)}{d\xi} + c \frac{d^3 f(\xi)}{d\xi^3} + g(\xi) \frac{dg(\xi)}{d\xi} = 0,$$

$$-c \frac{dg(\xi)}{d\xi} + g(\xi) \frac{dg(\xi)}{d\xi} + c \frac{d^3g(\xi)}{d\xi^3} = 0. \quad (28)$$

We define the two functions

$$f(\xi) = \lambda_1 \cos^{\beta_1}(\mu_1 \xi), \quad g(\xi) = \lambda_2 \cos^{\beta_2}(\mu_2 \xi). \quad (29)$$

Substituting Eq. (29) into Eq. (28) we obtain a system of algebraic equations

$$\begin{aligned} -c\lambda_1 - c\lambda_1\beta_1\mu_1^2 - c\lambda_1\mu_1\beta_1(\beta_1 - 1) &= 0, & \beta_1 + 2\beta_2 + 2 &= 0, \\ \frac{\lambda_1^2}{2} + \frac{\lambda_2^2}{2} + c\lambda_1\mu_1^2\beta_1(\beta_1 - 1) &= 0, & -c\lambda_2 - c\lambda_2\beta_2\mu_2^2 - c\lambda_2\mu_2\beta_2(\beta_2 - 1) &= 0, & \beta_2 + 2 &= 0, \\ \frac{\lambda_2^2}{2} + c\lambda_2\mu_2^2\beta_2(\beta_2 - 1) &= 0. \end{aligned} \quad (30)$$

Solving the system of Eq. (30), with the aid of MATHEMATICA, we have

$$\begin{aligned} \lambda_1 &= (0.5 - 2.95804i)c, & \beta_1 &= 2, & \mu_1 &= \pm 0.5i, \\ \lambda_2 &= 3c, & \beta_2 &= -2, & \mu_2 &= \pm 0.5i. \end{aligned} \quad (31)$$

So we obtain the exact soliton solution of the CEWE equation in the form

$$\begin{aligned} u(x, t) &= (0.5 - 2.95804i)c \cosh^2[0.5(x - ct)], \\ v(x, t) &= 3c \operatorname{sech}^2[0.5(x - ct)]. \end{aligned} \quad (32)$$

4. Conclusion

In this Letter, the cosine-function method has been successfully applied to find the solution for five nonlinear partial differential equations such as GEWE, GRLW, GKdV, GIKdV and CEWE equations. The cosine-function method is used to find a new exact solution. Thus, we can say that the proposed method can be extended to solve the problems of nonlinear partial differential equations which arising in the theory of solitons and other areas.

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References

- [1] A.R. Mitchell, D.F. Griffiths, The Finite Difference Method in Partial Equations, John Wiley & Sons, 1980.
- [2] E.J. Parkes, B.R. Duffy, Comput. Phys. Commun. 98 (1998) 288.
- [3] A.H. Khater, W. Malfiet, D.K. Callebaut, E.S. Kamel, Chaos Solitons Fractals 14 (2002) 513.
- [4] E. Fan, Phys. Lett. A 277 (2000) 212.
- [5] E. Fan, Z. Naturforsch. A 56 (2001) 312.
- [6] S.A. Elwakil, S.K. El-Labany, M.A. Zahran, R. Sabry, Phys. Lett. A 299 (2002) 179.
- [7] S.A. EL-Wakil, M.A. Abdou, Chaos Solitons Fractals 31 (2007) 840.
- [8] S.A. EL-Wakil, M.A. Abdou, Chaos Solitons Fractals 31 (2007) 1256.
- [9] S.A. EL-Wakil, M.A. Abdou, New exact traveling wave solution for two nonlinear physical models, Nonlinear Anal., in press.
- [10] A.A. Soliman, Physica A 361 (2) (2006) 394.
- [11] M.A. Abdou, A.A. Soliman, Phys. Lett. A 353 (2006) 487.
- [12] M.A. Abdou, A.A. Soliman, J. Comput. Appl. Math. 181 (2) (2005) 245.
- [13] A.A. Soliman, Math. Comput. Simul. 70 (2) (2005) 119.
- [14] M.A. Abdou, A.A. Soliman, Physica D 211 (2005) 1.
- [15] A.A. Soliman, Chaos Solitons Fractals 29 (2006) 294.
- [16] A.A. Soliman, M.A. Abdou, Numerical solutions of nonlinear evolution equations using variational iteration method, J. Comput. Appl. Math. (2007), in press.
- [17] M.A. Abdou, On the variational iteration method, Phys. Lett. A (2007), in press.
- [18] Y.T. Gao, B. Tian, Comput. Phys. Commun. 133 (2001) 158.
- [19] B. Tian, Y.T. Gao, Z. Naturforsch. A 57 (2002) 39.
- [20] X.-Y. Tang, S.-Y. Lou, Chaos Solitons Fractals 14 (2002) 1451.
- [21] X.-Y. Tang, S.-Y. Lou, Phys. Rev. E 66 (2002) 46601.
- [22] A.M. Wazwaz, Appl. Math. Comput. 161 (2) (2005) 561.
- [23] A.M. Wazwaz, Appl. Math. Comput. 161 (2) (2005) 575.
- [24] L.R.T. Gardner, G.A. Gardner, J. Comput. Phys. (1990) 218.
- [25] S.I. Zaki, Comput. Methods Appl. Mech. Engrg. 189 (2000) 587.
- [26] S.I. Zaki, Comput. Methods Appl. Mech. Engrg. 190 (2001) 4881.

- [27] K.R. Raslan, Numerical methods for partial differential equations, Ph.D. thesis, Al-Azhar University, Cairo, 1999.
- [28] K.R. Raslan, *Int. J. Comput. Math.* 81 (2004) 63.
- [29] D.J. Evans, K.R. Raslan, *Int. J. Comput. Math.* 81 (4) (2005) 445.
- [30] K.R. Raslan, *J. Simul. Process Modeling* 25 (2004).
- [31] J.C. Elibeck, G.R. McGuire, *J. Comput. Phys.* 23 (1977) 63.
- [32] L.R.T. Gardner, G.A. Gardner, *J. Comput. Phys.* 91 (1990) 441.
- [33] P.C. Jain, R. Shankar, T.V. Singh, *Commun. Numer. Methods Eng.* 9 (1993) 587.
- [34] L.R.T. Gardner, G.A. Gardner, A. Dogan, *Commun. Numer. Methods Eng.* 12 (1996) 795.
- [35] S.I. Zaki, *Comput. Phys. Commun.* 138 (2001) 80.
- [36] A.H.A. Ali, Finite element studies of the Korteweg–de Vries equation, Ph.D. thesis, University College of North Wales, Bangor, 1989.
- [37] Kh.O. Abdulloev, I.L. Bogolubsky, V.G. Makhankov, *Phys. Lett. A* 56 (6) (1976) 427.